

Decision Support Systems

IS 272

Course: IS 272 (Decision Support Systems)

Lecturer: Dr. Stephen Anokye

Assesment

- Attendance: 10%
- Quizes, Assignments : 30%
- Project : 40%
- End of Semester Examinations: 20%

- Virtually everyone makes hundreds of decisions each day.
- These decisions range from the inconsequential, such as what to eat for breakfast, to the significant, such as how best to get the economy out of a recession.
- All other things being equal, good outcomes from those decisions are better than bad outcomes.
- For most of us, however, good outcomes in decision making are a result of making good decisions.

- "Good decision making" means we are informed and have relevant and appropriate information on which to base our choices among alternatives on.
- In some cases, we support decisions using existing, historical data, while other times we collect the information, especially for a particular choice process.
- The information comes in the form of facts, numbers, impressions, graphics, pictures, and sounds.
- It needs to be collected from various sources, joined together, and organized.
- The process of organizing and examining the information about the various options is the process of modeling.
- Models are created to help decision makers understand the ramifications of selecting an option.
- The models can range from quite informal representations to complex mathematical relationships.

Discussion of decision to:

Decide what to eat for a meal?

Improving our customer relationships

- The quality of the decision depends on the adequacy of the available information, the quality of the information, the number of options, and the appropriateness of the modeling effort available at the time of the decision.
- While it is not true that more information (or even more analysis) is better, it is true that more of the appropriate type of information (and analysis) is better.
- In fact, one might say that to improve the choice process, we need to improve the information collection and analysis processes.

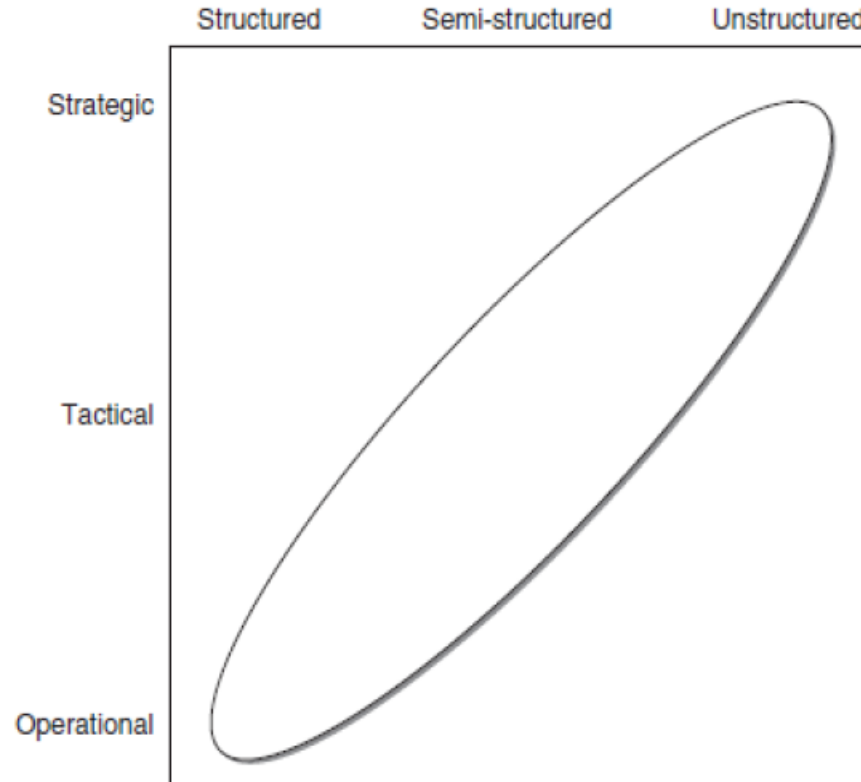
- Increasingly corporations are attempting to make more informed decisions to improve their bottom lines.
- Some refer to these efforts to use better information and better models to improve decision making as business intelligence. Others refer to it as analytics.
- In either case, the goal is to bring together the right information and the right models to understand what is going on in the business and to consider problems from multiple perspectives so as to provide the best guidance for the decision maker.
- One way to accomplish the goal of bringing together the appropriate information and models for informed decision making is to use decision support systems (DSS)

Types of decisions

- Unstructured
- Structured
- Semi structured

Types of Decisions

- **Decisions can be classified in terms of two main dimensions:**
 - Nature
 - Scope
- **According to Nature Decisions are classified into:**
 - Structured
 - Unstructured
 - Semi-Structured
- **According to Scope Decisions are classified into:**
 - Strategic
 - Tactical
 - Operational



A Taxonomy of Decisions

- Unstructured:

- Decision maker must provide judgment, evaluation, and insight to solve problem
- If the 3 phases of intelligence, design and choice are also unstructured

- Structured:

- Repetitive and routine; involve definite procedure for handling so they do not have to be treated each time as new
- If it is based on a well-defined and recurring decision making procedure

- Semi structured:

- Only part of problem has clear-cut answer provided by accepted procedure
- When some phases are structured and others are not

- Strategic :

- When they affect the entire organization or at least a substantial part of it for a long period of time.

Tactical :

- Affects only part of an enterprise and are usually restricted to a single department for a short period eg. a year

- Operational :

- Refer to specific activity carried out within an organization and have a modest impact on the future.

Decision Characteristics

Unstructured

Semistructured

Structured

Senior
Management

Middle Management

Operational Management
Individual Employees and Teams

Examples of Decisions

Decide entrance or exit
from markets
Approve capital budget
Decide long-term goals

Design a marketing plan
Develop a departmental
budget
Design a new corporate
Web site

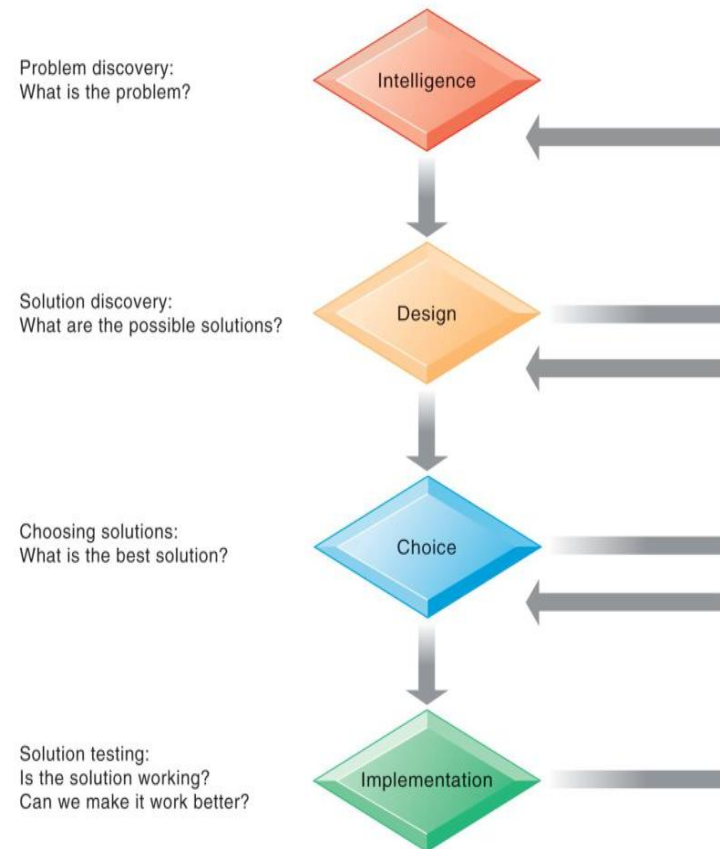
Determine overtime
eligibility
Restock inventory
Offer credit to customers
Determine special offers
to customers

Senior managers, middle managers, operational managers, and employees have different types of decisions and information requirements.

DECISION-MAKING & MIS

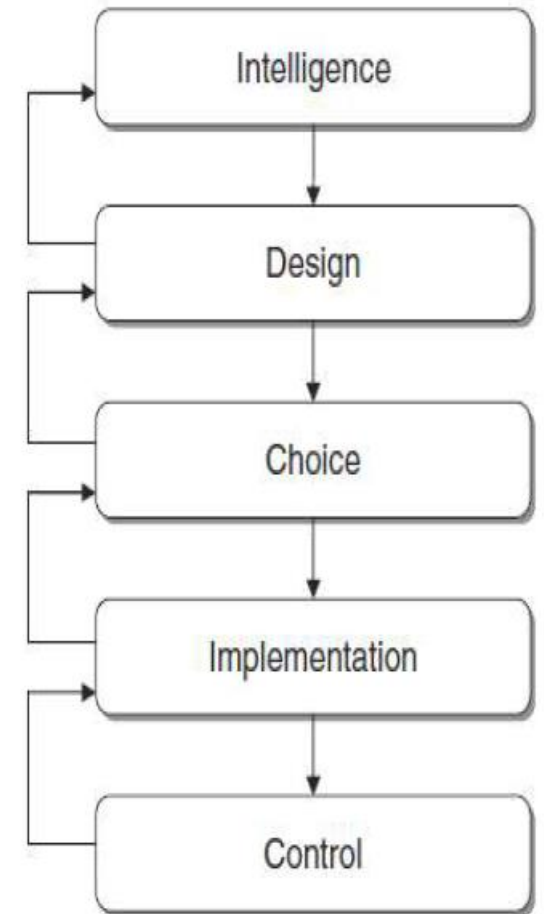
The four stages of the decision-making process

- Intelligence
- Design
- Choice
- Implementation



....in other literature

- This model includes 3 phases :
 - Intelligence
 - Design
 - Choice
- Two more phases are added and logical structure is then extended for the phases of Decision Making Process.
 - Two more phases are :
 - Implementation
 - Control



Phases of the decision-making process

The four stages of the decision-making process

1. Intelligence

- Discovering, identifying, and understanding the problems occurring in the organization
- In the intelligence phase the task of the decision maker is to identify, circumscribe and explicitly define the problem that emerges in the system under study.
- The analysis of the context and all the available information may allow decision makers to quickly grasp the signals and symptoms pointing to a corrective action to improve the system performance.

2. Design

- Identifying and exploring solutions to the problem
- In the design phase actions aimed at solving the identified problem should be developed and planned.
- At this level, the experience and creativity of the decision makers play a critical role, as they are asked to devise viable solutions that ultimately allow the intended purpose to be achieved.
- Where the number of available actions is small, decision makers can make an explicit enumeration of the alternatives to identify the best solution
- If, on the other hand, the number of alternatives is very large, or even unlimited, their identification occurs in an implicit way, usually through a description of the rules that feasible actions should satisfy

3. Choice

- Choosing among solution alternatives
- Once the alternative actions have been identified, it is necessary to evaluate them on the basis of the performance criteria deemed significant.
- Mathematical models and the corresponding solution methods usually play a valuable role during the choice phase
- Eg. Decision trees can be used to handle decision-making processes influenced by stochastic events.

4. Implementation

- Making chosen alternative work and continuing to monitor how well solution is working
- When the best alternative has been selected by the decision maker, it is transformed into actions by means of an implementation plan.
- This involves assigning responsibilities and roles
- to all those involved into the action plan.

5. Control

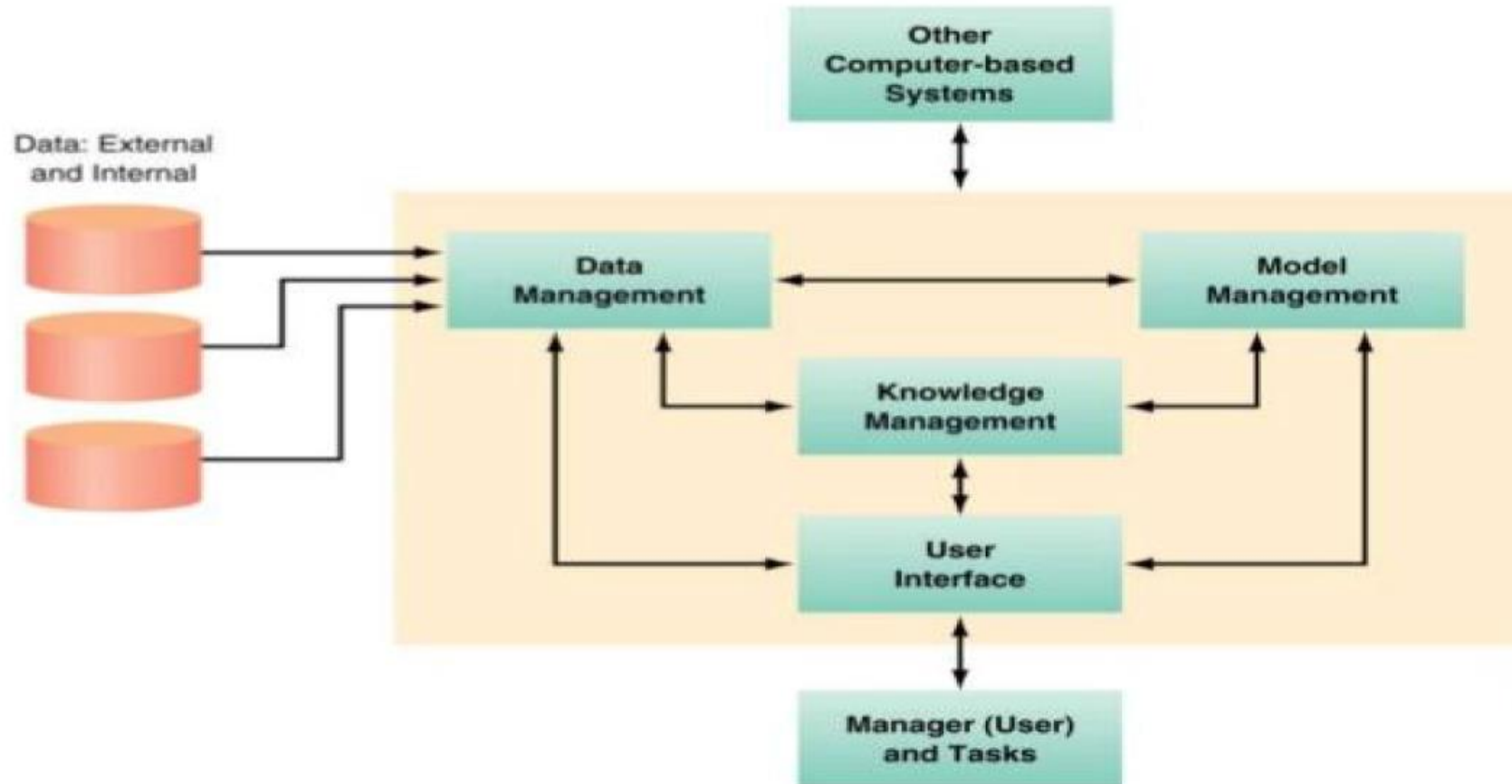
- Once the action has been implemented, it is finally necessary to verify and check that the original expectations have been satisfied and the effects of the action match the original intentions.
- In particular, the differences between the values of the performance indicators identified in the choice phase and the values actually observed at the end of the implementation plan should be measured.
- In an adequately planned DSS, the results of these evaluations translate into experience and information, which are then transferred into the data warehouse to be used during subsequent decision-making processes

Definition of DSS

- **Michael S. Scott Morton** has defined DSS as an interactive computer-based system that helps decision-makers utilize data and models to solve unstructured problems.
- **Gorry & Scott Morton** defined DSS as a system to support managerial decision-makers in semi-structured and unstructured decision situations.
- **Little** has characterized DSS as a model-based set of procedures, data and judgment to assist the manager in his decision-making tasks.
- **Bonczek** characterized DSS as a computer-based interactive system consisting of three components. These three components are:
 - Language system to provide communication between user and other components of the system,
 - a knowledge system,
 - (a problem processing system).
- **Keen** has viewed DSS as a tool for providing access to information and analytic models directly to decision-makers in order to support decision tasks that blend the use of both intuitive judgment and analytical methods.

- Decision-support systems are used to make business decisions, often based on data collected by on-line transaction-processing systems.
- Examples of business decisions:
 - What items to stock?
 - What insurance premium to change?
 - whom to send advertisements?
- Examples of data used for making decisions
 - Retail sales transaction details
 - Customer profiles (income, age, gender, etc.)

Components and Structure of a DSS



Data Management

- Includes the database(s) containing relevant data for the decision situation

User Interface

- Enables the users to communicate with and command the DSS

Model Management

- Includes software with financial, statistical, management science, or other quantitative models

Knowledge Management

- Provides knowledge for solution of the problem; supports any of the other subsystems or act as an independent component

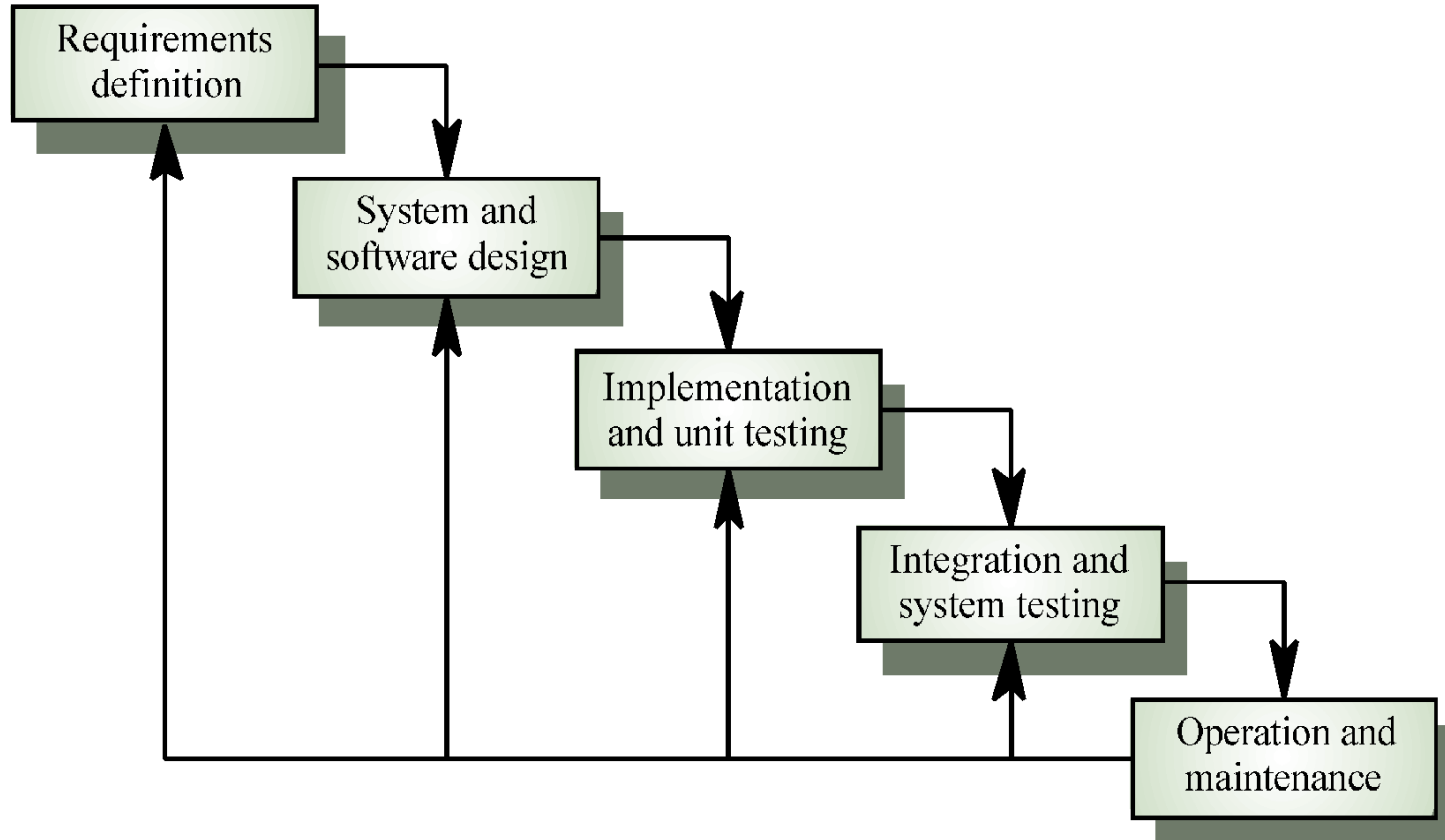
Why DSS?

- Increasing complexity of decisions
 - Technology
 - Information:
 - “Data, data everywhere, and not the time to think!”
 - Number and complexity of options
 - Pace of change
- Increasing availability of computerized support
 - Inexpensive high-powered computing
 - Better software
 - More efficient software development process
- Increasing usability of computers
- Consistency

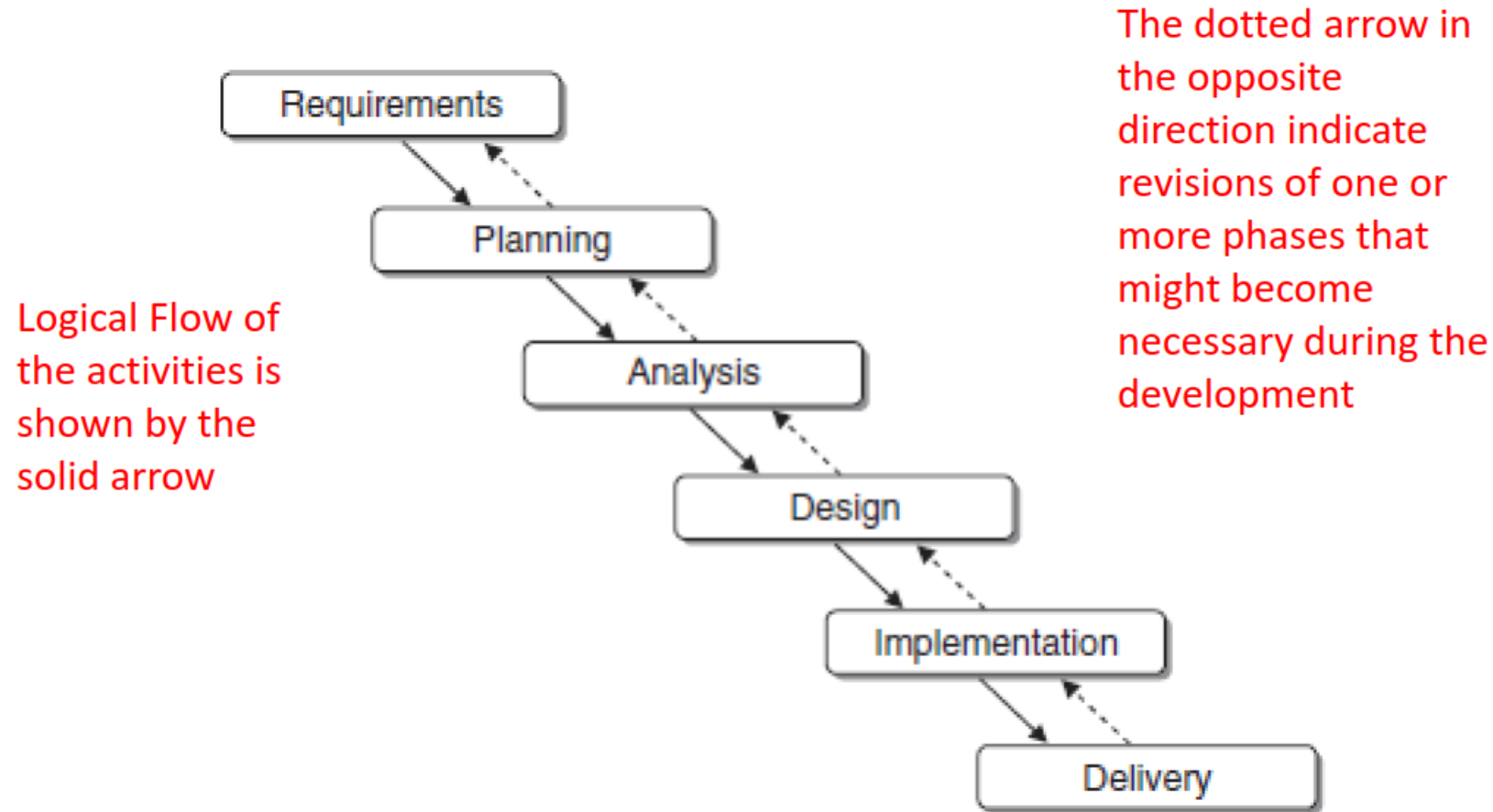
Perceived benefits of a DSS

- decision quality
- improved communication
- cost reduction
- increased productivity
- time savings
- improved customer and employee satisfaction

Waterfall Model



Development of a DSS



Phases in the development of a decision support system

- Planning
 - Why do we wish to develop a DSS?
- Analysis
 - What should the DSS accomplish, and who will use it; when and how?
- Design
 - How will the DSS work?
- Implementation
 - Change management
 - Rapid prototype development
 - Agile development technique
 - Extreme programming technique

Change Management

- Crucial to DSS
- People resistant to change
- Examine cause of change
- May require organizational culture shift
- Lewin-Schein change theory: three steps
 - Unfreeze
 - ✓ Create awareness of need for change
 - ✓ People support what they help create
 - Move
 - ✓ Develop new methods, attitudes and behaviors
 - ✓ Create and maintain momentum
 - Refreeze
 - ✓ Reinforce desired changes
 - ✓ Establish stable environment

Characteristics of a DSS

- A DSS must access data from a variety of sources.
- It facilitates the development and evaluation of a model of the choice process. That is, the DSS must allow users to transform the enormous amount of "data" into "information" which helps them make a good decision.
- It must provide a good user interface through which users can easily navigate and interact.
- Support decision makers at all managerial levels
- Support several interdependent and/or sequential decisions
- Support all phases of decision making and a variety of decision-making processes and styles
- Can be adapted over time to deal with changing conditions
- Easy to construct
- Utilizes models and links to data- and knowledge bases
- Execute sensitivity analysis

DSS Analysis

Sensitivity Analysis

- The study of the effect that changes in one or more parts of a model have on other parts of the model

What-if Analysis

- Checks the impact of a change in the assumptions or other input data on the proposed solution

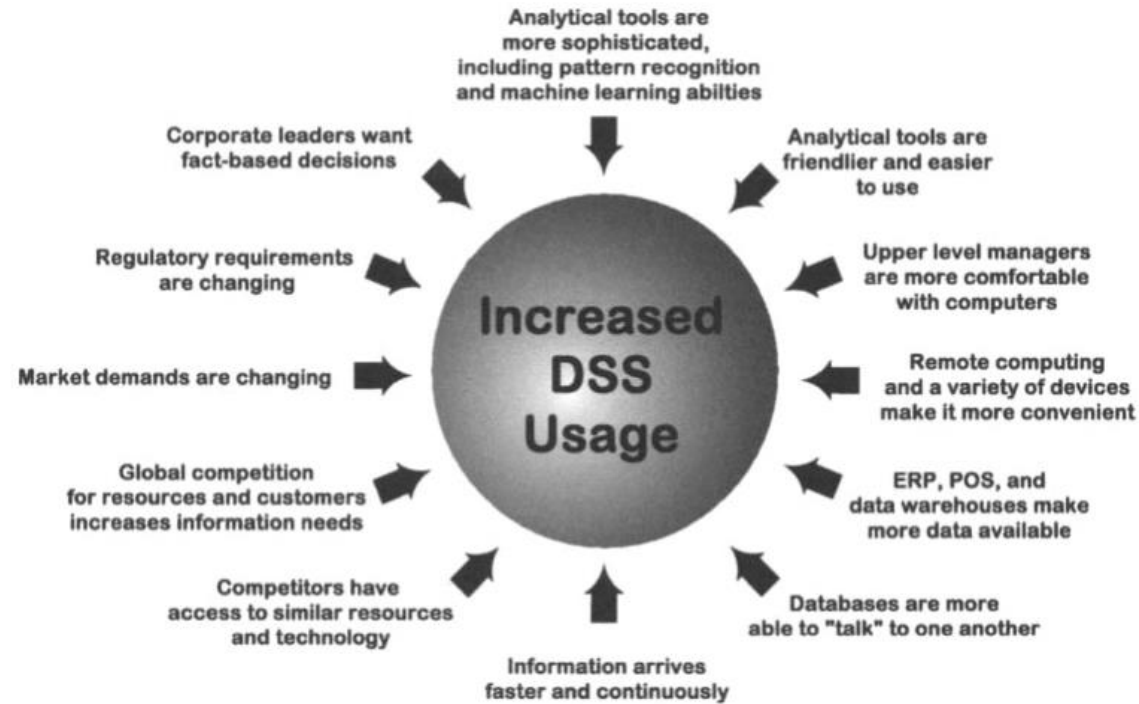
Goal-seeking Analysis

- Find the value of the inputs necessary to achieve a desired level of output

Optimisation Analysis

- Identifying the optimum level of output

Factors that influence businesses to use DSS



Factors that may affect the degree of success of DSS

- Integration: The design and development of a DSS require a significant number of methodologies, tools, models, individuals and organizational processes to work in harmony.
- Involvement: The exclusion or feeling of isolation from the project team of knowledge workers who will actually use the system once it is implemented is a mistake that is sometimes made during the design and development of DSS.
- Uncertainty: Implementation cost is less but driving more effective decisions may cost more.

DSS Classification-Relationship with the User

Passive DSS

- System that aids the process of decision making, but that cannot bring out explicit decision suggestions or solutions.

Active DSS

- Can bring out such decision suggestions or solutions.

Cooperative DSS

- Allows the decision maker to modify, complete, or refine the decision suggestions provided by the system, before sending them back to the system for validation. The system again improves, completes, and refines the suggestions of the decision maker. Repeat this process.

DSS Classification-Mode of Assistance

Data-driven DSS

- Emphasizes access to and manipulation of a time series of internal company data and, sometimes, external data.

Model-driven DSS

- Use data and parameters provided by users to assist decision makers in analyzing a situation.

Communication-driven DSS

- Supports more than one person working on a shared task. Eg:- Microsoft's NetMeeting or Groove

Document-driven DSS

- Manages, retrieves and manipulates unstructured information in a variety of electronic formats.

Knowledge-driven DSS

- Provides specialized problem solving expertise stored as facts, rules, procedures

Business Intelligence in an Enterprise

Business intelligence

- Infrastructure for collecting, storing, analyzing data produced by business
 - Databases, data warehouses, data marts

Business analytics

- Tools and techniques for analyzing data
 - OLAP, statistics, models, data mining

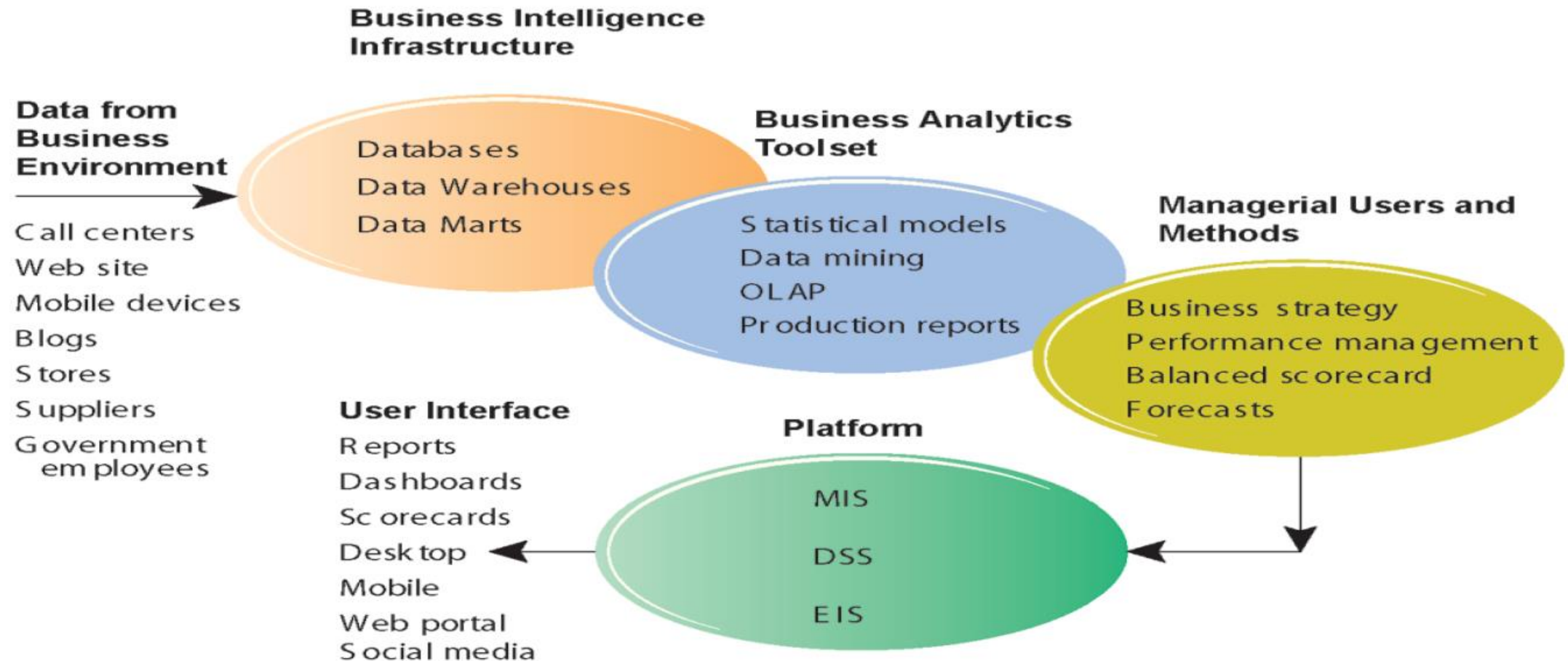
Business intelligence vendors

- Create business intelligence and analytics purchased by firms

Big data Analytics

- Massive datasets collected from social media, online and in-store customer data, and so on

Business Intelligence and Analytics for DSS



Business Intelligence in the Enterprise

- Six elements in the business intelligence environment
 - Data from the business environment
 - Business intelligence infrastructure
 - Business analytics toolset
 - Managerial users and methods
 - Delivery platform—MIS, DSS, ESS
 - User interface

Group Decision Support System (GDSS)

- Interactive system to facilitate solution of unstructured problems by group
- Specialized hardware and software; typically used in conference rooms
 - Overhead projectors, display screens
 - Software to collect, rank, edit participant ideas and responses
 - May require facilitator and staff
- Enables increasing meeting size and increasing productivity
- Promotes collaborative atmosphere, anonymity
- Uses structured methods to organize and evaluate ideas

Types of DSS

- Spreadsheets
- Relational Databases
- Artificial Intelligence
 - Expert Systems
 - case-based reasoning
 - fuzzy logic
 - Neural networks and data mining
 - Genetic algorithms
 - Intelligent agents

Spreadsheet

- BMI example
- Data inputs
- Data outputs
- Formulas
- Analysis
- Shortcomings

Artificial Intelligence

- **Waterman(1986):** Artificial intelligence was “the part of computer science concerned with developing intelligent computer programs.”
- **Rich (1983,):** “Artificial intelligence (AI) is the study of how to make computers do things at which, at the moment, people are better,”.
- **Winston (1992):** The study of computations that make it possible to perceive, reason and act.
- **Haugeland (1985):** The exciting new efforts to make computers think . . . machines with minds, in full and literal sense.
- **Kurzweil (1990):** The art of creating machines that can perform functions that require intelligence, when performed by people.
- **Stubblefield (1993):** The branch of computer science that is concerned with the automation of intelligent behavior.

Artificial Intelligence

- AI system falls into these basic categories:
 - Expert systems - knowledge based;
 - Natural Language (Native languages) Systems;
 - Perception System (vision, speech, touch);
 - Planning and Robotics
 - Machine Learning
- AI is a software technique applied to the nonnumeric data.
- The data is presented in symbols, statements, and patterns.
- AI uses the following for problem solving:
 - symbolic processing,
 - social and scientific reasoning,
 - Conceptual modeling.

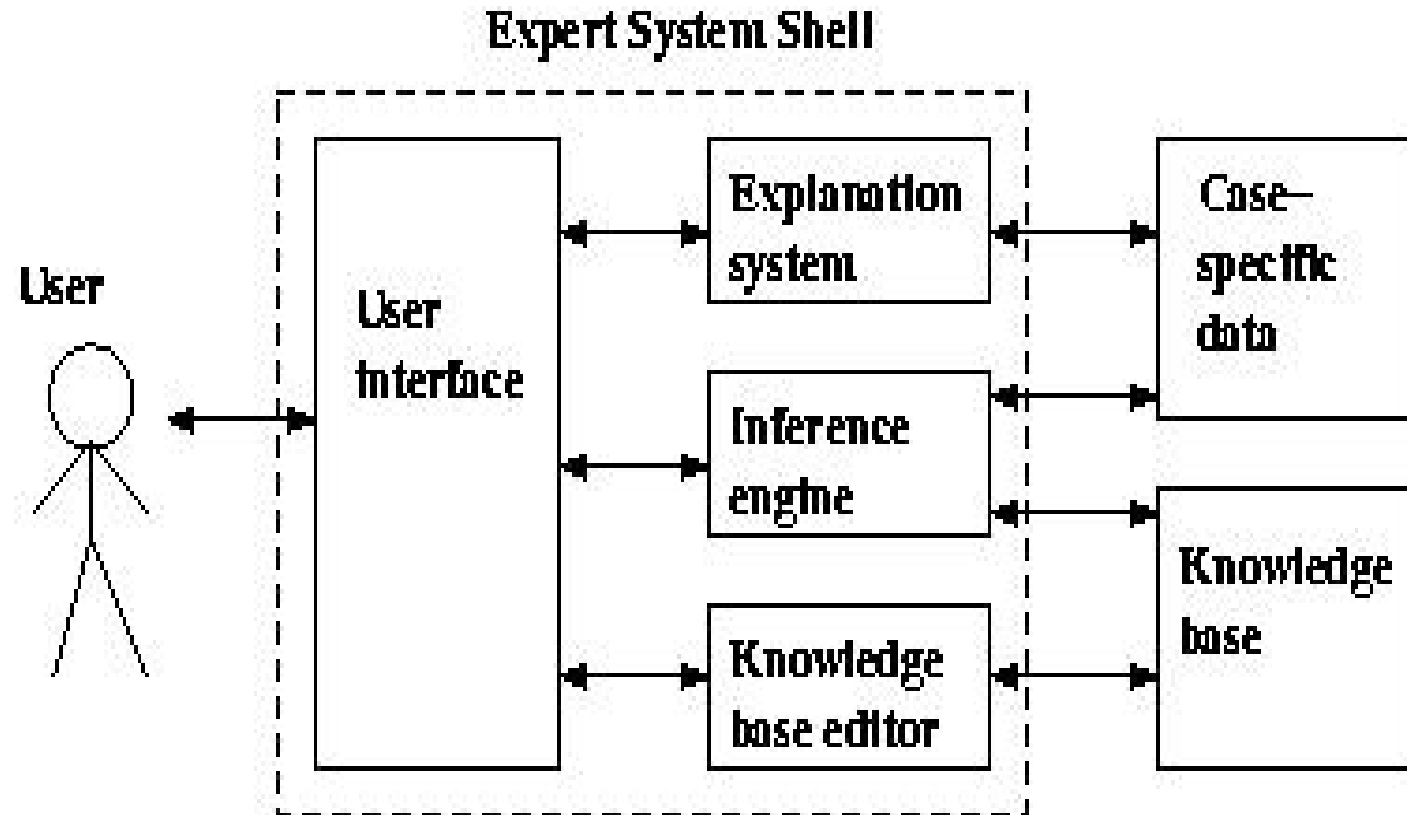
Expert Systems

- An expert system is a computer program that contains some of the subject-specific knowledge of one or more human experts.

Expert Systems

- Capture tacit knowledge in very specific and limited domain of human expertise
- Capture knowledge of skilled employees as set of rules in software system that can be used by others in organization
- Typically perform limited tasks that may take a few minutes or hours, for example:
 - Diagnosing malfunctioning machine
 - Determining whether to grant credit for loan
- Used for discrete, highly structured decision making

Expert System Architecture



- The user: interacts with the system through a *user interface* which may use menus, natural language or any other style of interaction).
- Then *inference engine*: is used to reason with both the *expert knowledge* (extracted from an expert) and data specific to the particular problem being solved.
- The expert knowledge: will typically be in the form of a set of IF-THEN rules.
- The *case specific data*: includes both data provided by the user and partial conclusions (along with certainty measures) based on this data.
- In a simple forward chaining rule-based system the case specific data will be the elements in *working memory*.
- *explanation subsystem* allows the program to explain its reasoning to the user.
- *knowledge base editor* helps the expert or knowledge engineer to easily update and check the knowledge base.

- The general purpose bit (in the dotted box in the figure) is referred to as an *expert system shell*.
- Expert systems work in industry includes both writing expert system shells and writing expert systems using shells.
- Using shells to write expert systems generally greatly reduces the cost and time of development (compared with writing the expert system from scratch).

- Expert systems tasks:
 - interpretation - inferring situation descriptions from sensor data
 - prediction - inferring likely consequences of given situations
 - diagnosis - inferring malfunctions from observations
 - prescription - prescribing remedies for malfunctions
 - design - configuring objects under constraints
 - planning - designing actions
 - monitoring - comparing observations to expected outcomes
 - control - governing overall system behaviour
 - instruction - diagnosing, prescribing and guiding users' behaviour

How Expert Systems Work

- Knowledge base: Set of hundreds or thousands of rules
- Inference engine: Strategy used to search knowledge base
 - Forward chaining: Inference engine begins with information entered by user and searches knowledge base to arrive at conclusion
 - Backward chaining: Begins with hypothesis and asks user questions until hypothesis is confirmed or disproved

Human Expertise	Artificial Expertise
i. Perishable ii. Difficult to transfer iii. Difficult to document iv. Unpredictable v. Expensive vi. Illogical vii. Not always diligent viii. Not always accessible ix. Not always available	i. Permanent (memory) ii. Easy to transfer iii. Easy to document iv. Consistent v. Affordable vi. Logical vii. Diligence viii. Persistent (accessibility) ix. Availability (multiple users at a time) x. longevity

Problems and Limitations of Expert Systems

- Knowledge is not always readily available
- Expertise can be hard to extract from humans
- Each expert's approach may be different, yet correct
- Hard, even for a highly skilled expert, to work under time pressure
- Expert system users have natural cognitive limits
- ES work well only in a *narrow domain* of knowledge
- Emotionless
- “Dumb”
- Predictable/Boring

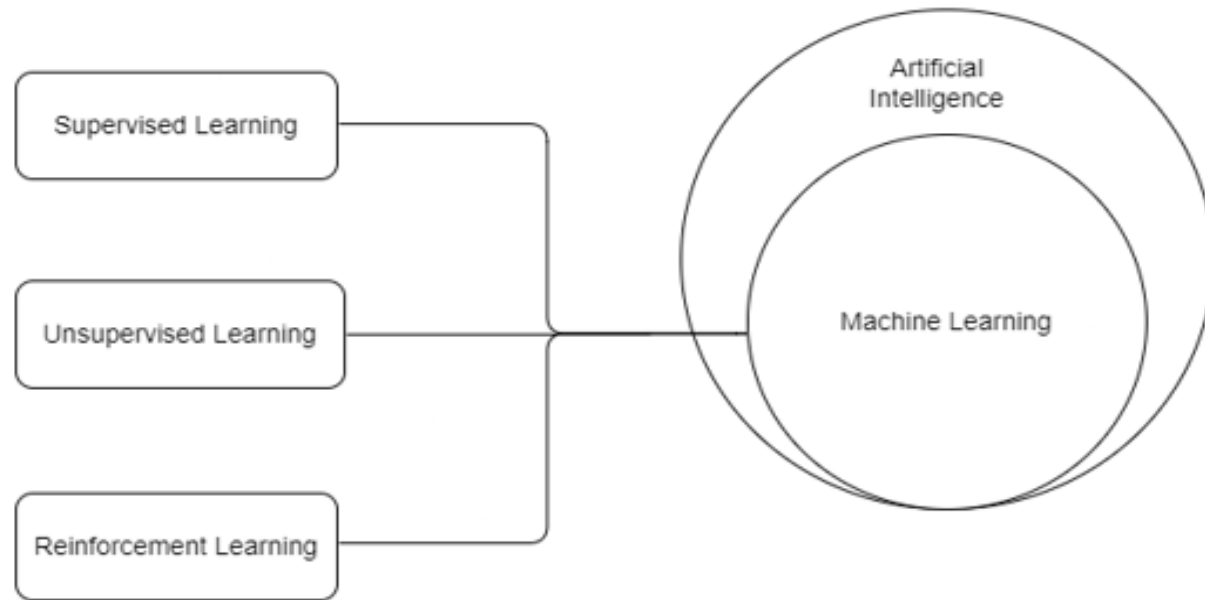
Machine Learning

- How computer programs improve performance without explicit programming
 - Recognizing patterns
 - Experience
 - Prior learnings (database)
- Contemporary examples
 - Google searches
 - Recommender systems on Amazon, Netflix

Machine Learning

- Machine learning (ML) is a field of study in artificial intelligence concerned with the development and study of statistical algorithms that can learn from data and generalize to unseen data and thus perform tasks without explicit instructions
- Machine learning is about learning from data and making predictions and/or decisions.
- Usually we categorize machine learning as
 - supervised,
 - unsupervised, and
 - reinforcement learning.

Types Of Machine Learning



- Supervised Learning
 - Labelled data
 - Classification and Regression
- Unsupervised Learning
 - Unlabelled data
 - Clustering and Density estimation
- Supervised and unsupervised learning are usually one-shot, myopic, considering instant reward; while reinforcement learning is sequential, far-sighted, considering long-term accumulative reward.

Machine learning models and their training algorithms

Supervised learning	Unsupervised learning	Semi-supervised learning	Reinforcement learning
Data scientists provide input, output and feedback to build model (as the definition). EXAMPLE ALGORITHMS: Linear regressions ■ Sales forecasting. ■ Risk assessment. Support vector machines ■ Image classification. ■ Financial performance comparison. Decision trees ■ Predictive analytics. ■ Pricing.	Use deep learning to arrive at conclusions and patterns through unlabeled training data. EXAMPLE ALGORITHMS: Apriori ■ Sales functions. ■ Word associations. ■ Searcher. K-means clustering ■ Performance monitoring. ■ Searcher intent. Artificial neural networks ■ Generate new, synthetic data. ■ Data mining and pattern recognition.	Builds a model through a mix of labeled and unlabeled data, a set of categories, suggestions and example labels. EXAMPLE ALGORITHMS: Generative adversarial networks ■ Audio and video manipulation. ■ Data creation. Self-trained Naïve Bayes classifier ■ Natural language processing.	Self-interpreting but based on a system of rewards and punishments learned through trial and error, seeking maximum reward. EXAMPLE ALGORITHMS: Q-learning ■ Policy creation. ■ Consumption reduction. Model-based value estimation ■ Linear tasks. ■ Estimating parameters.

Artificial Neural Network (ANN)

- Find patterns and relationships in massive amounts of data too complicated for humans to analyze
- “Learn” patterns by searching for relationships, building models, and correcting over and over again
- Humans “train” network by feeding it data inputs for which outputs are known, to help neural network learn solution by example
- Used in medicine, science, and business for problems in pattern classification, prediction, financial analysis, and control and optimization

How Neural Networks work

- A neural network uses rules it “learns” from patterns in data to construct a hidden layer of logic.
- The hidden layer then processes inputs, classifying them based on the experience of the model.
- As an example, the neural network can be trained to distinguish between valid and fraudulent credit card purchases

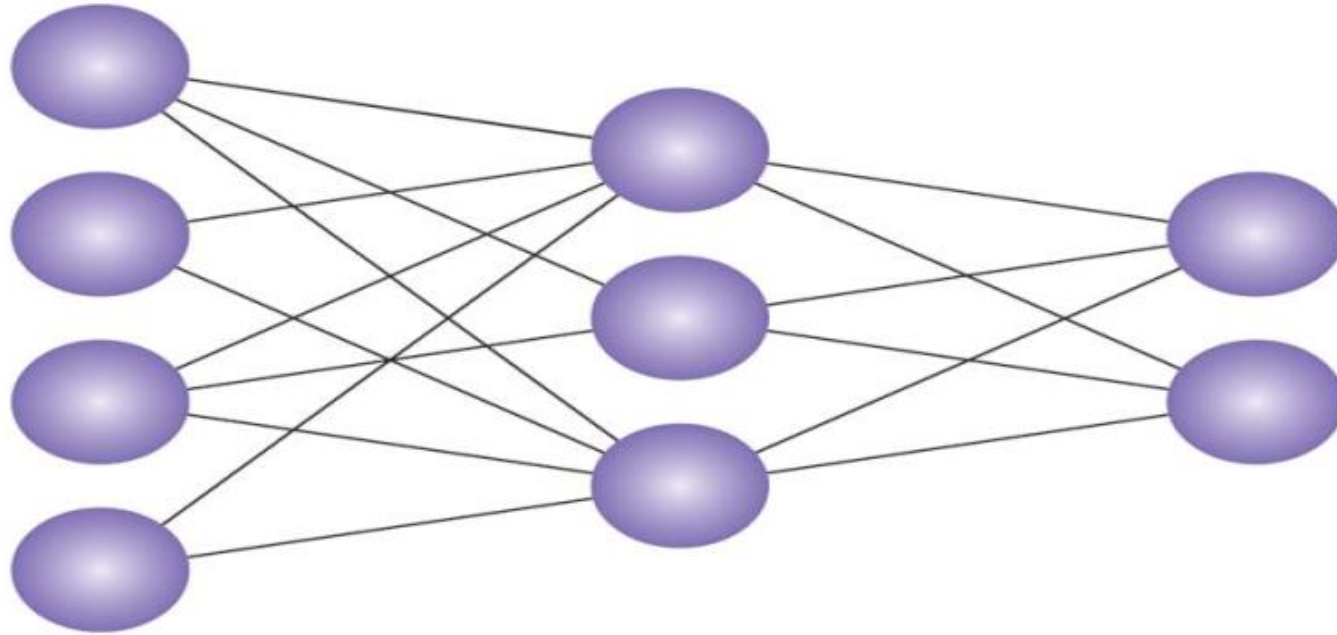
Data

- Age
- Income
- Purchase history
- Frequency of purchases
- Average purchase size

Input Layer

Hidden Layer

Output Layer



Results

Valid purchase

Fraudulent purchase

Fuzzy Logic

- Rule-based technology that represents imprecision used in linguistic categories (e.g., “cold,” “cool”) that represent range of values
- Describe a particular phenomenon or process linguistically and then represent that description in a small number of flexible rules
- Provides solutions to problems requiring expertise that is difficult to represent with IF-THEN rules
 - Autofocus in cameras
 - Detecting possible medical fraud

Genetic Algorithms

- Useful for finding optimal solution for specific problem by examining very large number of possible solutions for that problem
- Conceptually based on process of evolution
 - Search among solution variables by changing and reorganizing component parts using processes such as inheritance, mutation, and selection
- Used in optimization problems (minimization of costs, efficient scheduling, optimal jet engine design) in which hundreds or thousands of variables exist
- Able to evaluate many solution alternatives quickly

Hybrid AI Systems

- Genetic algorithms, fuzzy logic, neural networks, and expert systems integrated into single application to take advantage of best features of each
- For example: Matsushita “neurofuzzy” washing machine that combines fuzzy logic with neural networks

Intelligent Agents

- Work without direct human intervention to carry out specific, repetitive, and predictable tasks for user, process, or application
 - Deleting junk e-mail
 - Finding cheapest airfare
- Use limited built-in or learned knowledge base
 - Some are capable of self-adjustment, for example: Siri
- Agent-based modeling applications:
 - Systems of autonomous agents
 - Model behavior of consumers, stock markets, and supply chains; used to predict spread of epidemics